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Software Requirements Specification (SRS)

E-Commerce Website (Amazon-like Platform)

Document Version: 1.0 **Purpose:** This document defines functional and non-functional requirements for an Amazon-style e-commerce platform, intended to serve as a base for QA test case generation.

1. Overview

The system is a multi-vendor/single-vendor e-commerce website enabling customers to browse products, add them to a cart, place orders, make payments, and track deliveries. It includes customer-facing modules, a seller/admin panel, and backend services for inventory, payments, and order fulfillment.

User Roles: - **Guest User** - Unregistered visitor - **Registered Customer** - Logged-in buyer
- **Seller/Vendor** - Lists and manages products - **Admin** - Manages platform-wide operations
- **Delivery Partner** - Manages shipment/delivery status (optional module)

2. Functional Requirements

2.1 User Registration & Authentication

- FR-1.1: User shall be able to register using email/phone number and password.
- FR-1.2: System shall validate email format and enforce password strength rules (min 8 chars, 1 uppercase, 1 number, 1 special character).
- FR-1.3: User shall receive an OTP/email verification link upon registration.
- FR-1.4: User shall be able to log in using email/phone + password.
- FR-1.5: System shall support social login (Google, Facebook, Apple).
- FR-1.6: User shall be able to reset password via “Forgot Password” using OTP/email link.
- FR-1.7: System shall lock account after 5 consecutive failed login attempts for 15 minutes.
- FR-1.8: User shall be able to log out from all/current devices.
- FR-1.9: System shall support two-factor authentication (optional, toggleable in account settings).
- FR-1.10: Guest users shall be able to browse and add to cart without login, but must log in/register at checkout.

2.2 User Profile & Account Management

- FR-2.1: User shall be able to view/edit profile (name, email, phone, DOB, gender).
- FR-2.2: User shall be able to add, edit, delete, and set default multiple shipping addresses.
- FR-2.3: User shall be able to add/manage multiple payment methods (cards, UPI, wallets).
- FR-2.4: User shall be able to view order history and order status.
- FR-2.5: User shall be able to deactivate/delete account.
- FR-2.6: User shall be able to manage notification preferences (email/SMS/push).

2.3 Product Catalog & Browsing

- FR-3.1: System shall display products organized by categories and subcategories.
- FR-3.2: User shall be able to search products by keyword, brand, or category.
- FR-3.3: Search shall support autosuggestions/autocomplete.
- FR-3.4: System shall support search filters: price range, brand, ratings, discount, availability, size, color.
- FR-3.5: System shall support sorting: price (low-high/high-low), popularity, newest, customer rating.
- FR-3.6: Each product listing shall display: image(s), title, price, discount, rating, delivery estimate, stock status.
- FR-3.7: Product detail page shall show: description, specifications, multiple images/zoom, price, variants (size/color), seller info, ratings & reviews, Q&A section.
- FR-3.8: System shall display “out of stock” status and disable purchase actions accordingly.
- FR-3.9: System shall show related/recommended products (“Customers also bought”, “Similar items”).
- FR-3.10: System shall support product comparison (select 2-4 products to compare specs).

2.4 Shopping Cart

- FR-4.1: User shall be able to add products to cart with selected quantity/variant.
- FR-4.2: User shall be able to update quantity or remove items from the cart.
- FR-4.3: Cart shall persist across sessions for logged-in users.
- FR-4.4: Cart shall display item-wise price, total price, taxes, shipping estimate, and discounts.
- FR-4.5: System shall validate stock availability before checkout.
- FR-4.6: User shall be able to apply/remove coupon codes in the cart.
- FR-4.7: System shall recalculate totals dynamically when cart contents change.
- FR-4.8: User shall be able to save items “for later” / move to wishlist from cart.

2.5 Wishlist

- FR-5.1: User shall be able to add/remove products to/from wishlist.
- FR-5.2: User shall be able to move wishlist items to cart.
- FR-5.3: Wishlist shall persist across sessions.

2.6 Checkout & Order Placement

- FR-6.1: User shall select/enter a shipping address during checkout.
- FR-6.2: User shall select a delivery option (standard/express) with estimated delivery date.

- FR-6.3: User shall select a payment method (Card, UPI, Net Banking, Wallet, COD).
- FR-6.4: System shall display an order summary (items, price breakup, taxes, discounts, delivery charges, total) before confirmation.
- FR-6.5: System shall validate payment details and process transaction securely.
- FR-6.6: System shall generate a unique Order ID upon successful payment/order placement.
- FR-6.7: System shall send order confirmation via email/SMS/push notification.
- FR-6.8: System shall handle payment failure gracefully with retry option.
- FR-6.9: System shall support Cash on Delivery (COD) where applicable, with eligibility checks.
- FR-6.10: System shall support order placement for multiple sellers in a single checkout (split into sub-orders).

2.7 Order Management & Tracking

- FR-7.1: User shall be able to view list of all past and current orders.
- FR-7.2: User shall be able to view detailed order status (Placed, Confirmed, Shipped, Out for Delivery, Delivered, Cancelled, Returned).
- FR-7.3: User shall be able to track shipment with real-time status/tracking ID.
- FR-7.4: User shall be able to cancel an order before it is shipped.
- FR-7.5: User shall be able to request return/refund within the return policy window.
- FR-7.6: System shall process refunds to original payment method within a defined SLA.
- FR-7.7: User shall be able to download invoice for completed orders.
- FR-7.8: System shall notify users of order status changes.

2.8 Ratings & Reviews

- FR-8.1: User shall be able to rate (1-5 stars) and review a product only after purchase/delivery.
- FR-8.2: User shall be able to upload images/videos with reviews.
- FR-8.3: Other users shall be able to mark reviews as “helpful”.
- FR-8.4: System shall display average rating and rating distribution on product page.
- FR-8.5: User shall be able to edit/delete their own review.
- FR-8.6: System shall support reporting/flagging inappropriate reviews.

2.9 Payment & Pricing

- FR-9.1: System shall support multiple currencies/tax rules based on region (if applicable).
- FR-9.2: System shall calculate applicable taxes (GST/VAT) per item/order.
- FR-9.3: System shall integrate with third-party payment gateways securely (PCI-DSS compliant).
- FR-9.4: System shall support EMI/installment options on eligible cards.
- FR-9.5: System shall support gift cards and store credit.
- FR-9.6: System shall apply promotional discounts, coupon codes, and bank offers correctly at checkout.

2.10 Notifications

- FR-10.1: System shall send transactional notifications (order, shipment, delivery, refund) via email/SMS/push.
- FR-10.2: System shall send promotional notifications based on user preference/opt-in.
- FR-10.3: User shall be able to view an in-app notification center.

2.11 Seller/Vendor Module

- FR-11.1: Seller shall be able to register and complete KYC verification.
- FR-11.2: Seller shall be able to add/edit/delete product listings with images, price, stock, and description.
- FR-11.3: Seller shall be able to manage inventory levels.
- FR-11.4: Seller shall be able to view and process incoming orders.
- FR-11.5: Seller shall be able to view sales analytics/dashboard.
- FR-11.6: Seller shall receive payouts based on defined settlement cycle.

2.12 Admin Panel

- FR-12.1: Admin shall be able to manage users (view/block/delete accounts).
- FR-12.2: Admin shall be able to manage sellers and approve/reject product listings.
- FR-12.3: Admin shall be able to manage categories, banners, and promotions.
- FR-12.4: Admin shall be able to view platform-wide sales and order reports.
- FR-12.5: Admin shall be able to manage coupons/discount campaigns.
- FR-12.6: Admin shall be able to handle escalations, refunds, and disputes.
- FR-12.7: Admin shall be able to moderate reviews/reported content.

2.13 Search Engine & Recommendations

- FR-13.1: System shall provide typo-tolerant, relevance-ranked search results.
- FR-13.2: System shall provide personalized recommendations based on browsing/purchase history.
- FR-13.3: System shall support voice search (optional/mobile app).

3. Non-Functional Requirements

Category	Requirement
Performance	Page load time <= 3 seconds under normal load; search results returned within 1 second
Scalability	System shall support 100,000+ concurrent users during peak (e.g., sale events)
Availability	99.9% uptime SLA
Security	Data encryption in transit (TLS 1.2+) and at rest; PCI-DSS compliance for payment data; OWASP Top 10 mitigations
Usability	Responsive design supporting desktop, tablet, and mobile browsers
Compatibility	Support for latest 2 versions of Chrome, Firefox, Safari, Edge
Accessibility	WCAG 2.1 AA compliance
Localization	Support multiple languages and currencies
Data Retention	Order history retained for minimum 7 years (compliance)
Disaster Recovery	RPO <= 1 hour, RTO <= 4 hours
Auditability	All admin/seller actions shall be logged with timestamp and user ID

4. Key Business Rules

- BR-1: Orders above a defined threshold may require additional address/payment verification (fraud check).
 - BR-2: Products cannot be purchased in quantities exceeding available stock.
 - BR-3: Coupons shall have defined validity periods, usage limits, and minimum order value conditions.
 - BR-4: Returns are only allowed within the product-specific return window (e.g., 7/10/30 days), and some categories (e.g., perishables, innerwear) may be non-returnable.
 - BR-5: COD may be restricted based on pin code, order value, or user's COD order history.
 - BR-6: A user review is permitted only once per purchased product/order.
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5. Assumptions & Constraints

- Payment gateway integrations (e.g., Razorpay, Stripe, PayPal) are third-party and assumed to be reliable per their SLAs.
 - Shipping/logistics integration is assumed via third-party courier APIs.
 - Inventory sync between seller panel and live catalog is near real-time (< 5 min lag acceptable).
 - The system does not currently support B2B bulk ordering (unless explicitly scoped).
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6. Suggested Test Case Categories (for QA reference)

Based on the above requirements, test cases should be generated covering:

1. **Authentication & Authorization** - registration, login, password reset, session handling, role-based access
 2. **Product Search & Filters** - keyword search, filter combinations, sorting, edge cases (no results, special characters)
 3. **Product Detail Page** - variant selection, stock validation, image gallery, reviews display
 4. **Cart Operations** - add/remove/update, quantity limits, coupon application, price recalculation
 5. **Checkout Flow** - address selection, payment method validation, order summary accuracy, payment failure handling
 6. **Order Management** - status transitions, cancellation rules, return/refund flow, invoice generation
 7. **Payment Processing** - successful/failed transactions, retries, refund timelines, multiple payment modes
 8. **Reviews & Ratings** - submission eligibility, edit/delete, media upload, moderation
 9. **Notifications** - trigger conditions, delivery channel accuracy, opt-in/opt-out behavior
 10. **Seller Panel** - listing management, inventory sync, order fulfillment
 11. **Admin Panel** - user/seller management, content moderation, reporting accuracy
 12. **Security** - SQL injection, XSS, CSRF, session hijacking, payment data handling
 13. **Performance & Load** - concurrent users, peak sale traffic, response times
 14. **Cross-Browser/Device** - responsive behavior across browsers and screen sizes
 15. **Negative & Edge Cases** - empty cart checkout, expired coupons, out-of-stock during checkout, network interruption during payment
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This document is intended as a functional baseline for test case design and may need refinement based on the actual product scope, target market, and technical architecture.